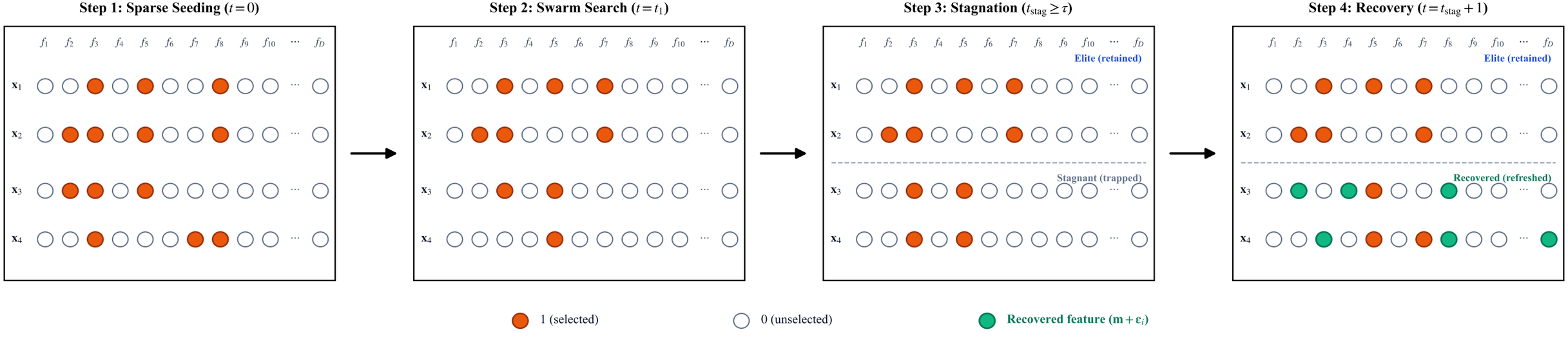
Step 1: Sparse Seeding ($t = 0$)
Step 2: Swarm Search ($t = t_1$)
Step 3: Stagnation ($t_{stag} \geq \tau$)
Step 4: Recovery ($t = t_{stag} + 1$)
$f_1$ $f_2$ $f_3$ $f_4$ $f_5$ $f_6$ $f_7$ $f_8$ $f_9$ $f_{10}$ ⋯ $f_D$
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}_3$
$\mathbf{x}_4$
Elite (retained)
Stagnant (trapped)
Recovered (refreshed)
1 (selected)
0 (unselected)
Recovered feature ($\mathbf{m} + \varepsilon_i$)